Neutrino oscillations as weight-preserving quantum circuits: what the qg symmetry filter reads, and what no oscillation circuit can see

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Abstract

A neutrino is produced and detected in a flavour state and propagates in mass states, so flavour oscillation is a rotation. We write it in the measurement-level units of the open-source library qang, where a qubit is described by qg_Z = <Z>, and place every flavour on its own qubit with a single excitation (the weight-1 sector). Every gate of the oscillation circuit then conserves the Hamming weight, and the qg symmetry filter (discard the shots that left the sector) applies. Six results follow, all on the noise models of IBM devices, each pre-registered and reported as passed or failed. (i) In vacuum $qg_Z = \cos^2 2\theta + \sin^2 2\theta \cos \Delta$ is the spherical law of cosines, and at the oscillation maximum the mixing angle is a quarter of the qang angle $\arccos(qg_Z)$, with the θ-independent error bar $1/(4\sqrt{(N+½)})$. (ii) Three flavours on three qubits: the filter cuts the error of the transition probabilities about 3× and the CP asymmetry at a DUNE-like baseline survives. (iii) Earth matter enters as a weight-preserving Trotter circuit, and matter fakes CP violation. (iv) In dense neutrino gases the conserved number is the number of ν_2, not the flavour number, so the filter applies in the mass basis (2.2-2.8×). (v) For solar neutrinos the phases are lost before detection, so the whole Sun is one rotation and the filter, though it works, is not needed. (vi) The mass ordering is read correctly through matter on three device models; Majorana phases change no oscillation probability, and the transpiler deletes them; a sterile neutrino is a fourth qubit in the weight-1 sector, where the filter cuts the error 5.5×. The link between neutrinos and qang is modelling, not new physics: the oscillation is a rotation, and qang reads rotations.

1. Oscillation is a rotation

With two flavours on one qubit (|0⟩ = ν_e), going to the mass basis (RY(−2θ)), letting the mass states pick up their relative phase Δ = Δm²L/2E (RZ) and coming back (RY(2θ)) gives P(e→μ) = sin²2θ sin²(Δ/2) and

$$qg_Z = 1 - 2P = \cos^2(2\theta) + \sin^2(2\theta)\cos\Delta,$$

the spherical law of cosines: the Bloch vector precesses on a cone of half-angle 2θ about the mass axis, and the qang angle arccos(qg_Z) is the great-circle distance travelled. At the maximum (Δ = π) it equals 4θ exactly, so θ = arccos(qg_Z)/4 inherits the θ-independent shot-noise standard deviation 1/(4√(N+½)) of the arcsine transformation. With N = 1000 shots, 2000 repetitions gave 0.0077-0.0081 against 0.0079 predicted. Elsewhere the relation is not linear: qg removes the sin² only at the maximum.

2. Three flavours in the weight-1 sector

Three flavours go on three qubits, qubit k excited = flavour k. The PMNS matrix U = R23 U_δ R13 U_δ† R12 is three RBS (XX+YY) rotations and two phase gates, and propagation is one phase gate per qubit, so every gate conserves the weight. At L = 1300 km, 0.8-5 GeV, from ν_μ, the filter with readout correction lowered the mean error of the nine transition probabilities from 0.025-0.048 to 0.009-0.013 on FakeBrisbane, FakeTorino and FakeFez (median 3.4×; 18 two-qubit gates, 86-94% of the shots kept). The CP asymmetry at 2.6 GeV (exact 0.025) read 0.022-0.026 filtered; it also survives without the filter, because it is a difference of two nearly identical circuits and the common shrinkage cancels.

3. Earth matter and dense neutrino gases

Electrons add A = 2√2 G_F N_e E to the ν_e entry of the Hamiltonian. Alternating the exact vacuum step with a phase gate on the ν_e qubit (second-order Trotter) keeps every gate weight-preserving. At the DUNE-like baseline matter alone produces an asymmetry of 0.042, larger than the vacuum CP signal (0.025), and the filtered circuit sees it (0.029-0.035 at two Trotter steps, whose own value is 0.033). The filter gained 2.8× (median). One Trotter step already had less error than the noise, so there was no depth trade-off.

In a supernova the neutrino density is so high that neutrinos oscillate together. With the vacuum term, what is conserved is the number of ν_2, not the flavour number, so the filter applies to mass-basis read-outs. All seven pre-registered predictions failed for one reason: the noiseless Trotter error (0.095) exceeded the noise. Scored post hoc against the noiseless circuit, the mass-basis filter gained 2.2-2.8× and reweighting by known sector probabilities 1.6-1.7×; the flavour basis, where nothing is conserved, gained nothing beyond readout correction.

4. Solar neutrinos: the whole Sun is one rotation

A ν_e born at the solar core follows the heavier matter eigenstate out of the Sun above a few MeV, so P_ee falls from 0.54 to 0.30. The mass states lose their coherence on the way, so the read-out is in the vacuum mass basis and $P_{ee} = c13^4(\cos^2\theta12\ p_1 + \sin^2\theta12\ p_2) + s13^4$ is linear in qg_Z of the ν_2 qubit. A slab circuit (exact 2×2 propagators, weight-preserving) gained 1.9-2.3× with the filter; but since only one population is observed, the whole Sun compiles to the single rotation that loads it, which beats every slab circuit even without the filter. A quantum computer could add something only where coherence survives.

5. What oscillation circuits can and cannot see

Mass ordering. With $\Delta m^2 31 < 0$ (inverted ordering) matter enhances the anti-neutrino μ→e instead of the neutrino. On three device models the filtered circuits read the ordering correctly, with the right sign of the ν/ν̄ asymmetry; the values sit 0.010-0.033 high, but the common bias cancels in NO − IO (0.034-0.036, exact 0.037).

Majorana phases. U_M = U diag(1, e^{iα21/2}, e^{iα31/2}) is diagonal in the mass basis, commutes with the propagation phases and cancels in U_M D U_M†: no oscillation probability depends on it, in vacuum or matter ($4\cdot10^{-16}$). On all three device models the transpiler merges the Majorana phase gates with their inverses and outputs the same circuit. Neutrinoless double beta decay sees them: $m_{\beta\beta} = |\Sigma\ U_{ei}^2\ m_i| = 1.4$-$3.7$ meV (normal ordering) and 19-48 meV (inverted) for a massless lightest neutrino.

A sterile neutrino (3+1) is a fourth flavour that does not interact: one more qubit in the weight-1 sector. With hypothetical $\theta14 = \theta24 = 0.15$ and $\Delta m^2 41 = 1\ eV^2$ at 0.5 km, the filter cut the error of P(μ→μ) 5.5× (median) and found the oscillation dip at the right energy (0.4 GeV) on all three device models.

6. Discussion and limits

The link between neutrinos and qang is modelling: oscillation is a rotation, and qang reads rotations and filters on conserved numbers. That gives a θ-independent error bar for the mixing angle, a 2-5× noise filter wherever the circuit conserves a number (flavour in vacuum and matter, ν_2 in dense gases, the sterile sector), and a clear statement of what a circuit cannot see (Majorana phases). It does not remove the trigonometry in general, does not speed up anything, and every result here is on simulated device noise with fixed parameters, not a fit to data. Of the pre-registered predictions behind this paper, 4 of 4 (§151), 4 of 5 (§152), 0 of 7 (§154), 6 of 6 (§156) and 5 of 6 (§163) passed; all are reported.

Code and data: github.com/Viny2030/qang (qang ≥ 0.12, module qang.neutrino); the scripts examples/neutrino_*_qg.py with their pre-registration hashes; RESEARCH_NOTES §151-§156 and §163; notebooks notebooks/qang_neutrinos.ipynb (English) and qang_neutrinos_es.ipynb (Spanish).

Summary of results

§	question	result
151	vacuum, 3 flavours	filter 3.4×; θ error bar 1/(4√(N+½))
152	Earth matter	filter 2.8×; matter A_CP 0.042 seen
154	dense gases	mass-basis filter 2.2–2.8× (post hoc)
156	solar	one rotation; filter 1.9–2.3× not needed
163	ordering, Majorana, sterile	ordering read; Majorana invisible; sterile 5.5×

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